

Question:

An aircraft weighting 250 kN and a tricycle undercarriage lands at a vertical velocity of 3.7 m/s, such that the vertical and horizontal reactions on the main wheels are 1200 kN and 400 kN respectively; at this instant the nose wheel is 1.0m from the ground, as shown in Fig. If the moment of inertia of the aircraft about its CG is $5.65 \times 10^8 \text{ N s}^2 \text{ mm}$, determine: the inertia forces on the aircraft, the time taken for its vertical velocity to become zero and its angular velocity at this instant.

